

Aaron Modiyil Joseph

London, UK

aaronjoseph1135@gmail.com | +44 7367 656 528

LinkedIn: [linkedin.com/in/aaron-modiyil-joseph](https://www.linkedin.com/in/aaron-modiyil-joseph) | Personal Portfolio: aaronmj7.github.io

PROFILE

MSc Data Science and BSc Mathematics graduate with strong analytical, statistical, and quantitative modelling skills developed through academic research and professional experience. Experienced in analysing structured financial and operational data, performing reconciliations, ensuring data quality, and producing accurate management reports within fast-paced environments. Proficient in Python, SQL, R, Excel, Power BI, and statistical modelling, with experience in forecasting, machine learning, automation, and data visualisation. Institute and Faculty of Actuaries (IFoA) Student Member with CS1 sat in April 2026 (results pending), seeking graduate opportunities across actuarial, pricing, reserving, capital modelling, insurance analytics, catastrophe modelling, and risk analytics. Passionate about applying quantitative methods to solve insurance and financial risk problems while progressing towards Fellowship.

EDUCATION

MSc. Data Science with Advanced Research (Distinction), University of Hertfordshire, Hatfield, UK (Jan 2023 – Mar 2025)

- Developed advanced quantitative modelling, statistical analysis and programming skills using Python, SQL, Excel and Git to analyse large complex datasets and generate business insights.
- Thesis: Predicting Movie Ratings and Analysing Feature Importance Using Tags and Genres
- Relevant Modules: Statistical Modelling, Python, SQL, Data Analysis, Data Visualisation, Anomaly Detection, Time Series Analysis, Machine Learning, AI Ethics.

BSc. Mathematics (Distinction), St. Berchmans College, Kerala, India (Jun 2019 – Sept 2022)

- Strong foundation in Probability, Statistics, Economics, Hypothesis Testing, Calculus, Linear Algebra, Differential Equations, and Real Analysis.
- Attained rigorous quantitative reasoning and problem-solving skills applicable across economic, financial, and statistical data analysis.

PROFESSIONAL MEMBERSHIPS & CERTIFICATIONS

- Institute and Faculty of Actuaries (IFoA) – Student Member
- Sitting CS1 (Actuarial Statistics) – September 2026
- Microsoft Certified: Azure AI Engineer Associate
- Microsoft Certified: Azure AI Fundamentals

TECHNICAL SKILLS

- **Actuarial & Quantitative Analysis:** Probability & Statistics, Statistical Modelling, Regression Analysis, Time Series Analysis, Forecasting, Risk Analysis, Quantitative Modelling, Predictive Analytics, Trend Analysis.
- **Programming & Analytics:** Python (Pandas, NumPy, Scikit-Learn), SQL, R, Machine Learning, Exploratory Data Analysis, Feature Engineering, Model Evaluation.
- **Data Management & Reporting:** Excel (Pivot Tables, VBA, Power Query), Power BI, Tableau, KPI Reporting, Management Information (MI), Dashboard Development, Data Cleaning, Data Validation, Data Quality Assurance, Data Reconciliation.
- **Other Tools:** Microsoft Azure, GitHub, Jupyter Notebook, Microsoft Office, LaTeX.

PROFESSIONAL EXPERIENCE

Night Auditor and Guest Service Assistant | Holiday Inn Express (IHG), Watford, UK (June 2023 – Present)

- Reconcile daily financial and operational records, ensuring accuracy across revenue, occupancy, and transaction data.
- Audit discrepancies across systems, identifying root causes and resolving data inconsistencies.
- Produce financial and operational reports, supporting performance monitoring, trend analysis, and management decision-making.
- Investigate monthly, quarterly, and annual occupancy, revenue, and customer reviews using dashboards to identify patterns, anomalies, and business performance insights.
- Manage confidential customer and financial information with accuracy and professionalism.
- Collaborate with finance, management, housekeeping, and front-office teams to resolve issues and improve reporting.

accuracy and workflow efficiency.

- Automated welcome letter generation for IHG members using Excel and Word, incorporating tier-level benefits and alphabetical ordering, replacing a manual system.
- Operate within a highly regulated, accuracy-focused environment requiring strong attention to detail, analytical thinking, and adherence to reporting deadlines.

Mathematics Tutor | Universal Tuition Centre, Kerala, India

(May 2021 – Jan 2023)

- Deliver structured mathematics and statistics tuition, developing strong analytical and quantitative reasoning skills.
- Communicate complex concepts in a clear and structured manner, strengthening stakeholder communication skills.
- Handle multiple student schedules and learning objectives, demonstrating organisation and discipline.

PROJECTS

Financial Market Forecasting and Risk Analysis

- Prepared forecasting models using historical market data to analyse pricing trends and market behaviour.
- Conducted data validation, cleaning, anomaly detection, time-series analysis and data preprocessing to identify market trends and predictive factors.
- Examined model performance using statistical error metrics against baseline forecasting methods and sensitivity analysis to evaluate forecasting reliability.
- Built Power BI dashboards to communicate financial trends, model outputs, and decision-support insights.

Comparative Study of ARIMA and LSTM Models on Time Series Data

- Tested ARIMA and LSTM forecasting models on seasonal and volatile financial datasets.
- Analysed seasonality, volatility, and temporal trends in time-series data, contributing to a comprehensive report that enhanced understanding of market patterns.
- Inspected forecasting accuracy using statistical error metrics and visualised results.
- Critiqued model suitability across different volatility environments, highlighting the importance of selecting appropriate modelling approaches for risk forecasting.

Comparative Study of Clustering Models

- Assessed Gaussian Mixture Model and K-Means clustering models on a large rental dataset.
- Traced relationships between quantitative variables through exploratory data analysis.
- Uncovered meaningful groupings and unusual patterns within large datasets.
- Evaluated clustering performance using statistical and visual assessment techniques.

Predicting Movie Ratings and Analysing Feature Importance Using Tags and Genres

- Implemented predictive models in Python to analyse relationships between user-generated tags, genres, and movie ratings within a large-scale dataset.
- Performed exploratory data analysis and feature engineering on large datasets.
- Applied statistical analysis to identify rating patterns and user behaviour trends.
- Reviewed model performance using statistical error metrics and visual analysis.

VOLUNTEERING & ACTIVITIES

- Active committee member of the Physics, Astronomy and Mathematics Society, Sim Racing Society, and Anime Society at the University of Hertfordshire.
- Monitored events and collaborated with diverse teams to deliver successful student activities.
- Led the Rubik's Cube competition during college events, demonstrating leadership, organisation, and teamwork skills.
- Achieved first place in several mathematics competitions during school and university, demonstrating strong mathematical skills and problem-solving abilities.

ADDITIONAL INFORMATION

Languages: Fluent in *English*, Hindi, Malayalam.

Hobbies and Interests: Financial Markets | Personal Finance | Investing | Board Games | Formula 1 | Chess | Football | Cricket | Rubik's Cubes | Playing Keyboard | Movies | Music